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Matter in Our Surroundings



When we look around us, we notice a wide range of objects with varying shapes, sizes, and textures. Everything in this universe is made of material that scientists call "matter." Everything is matter, including the air we breathe, the food we eat, stones, clouds, stars, plants and animals and even a single drop of water or sand. As we look around, we can see that all of the things mentioned above occupy space and have mass. In other words, they have mass as well as volume.



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TOPIC 1

MATTER IN OUR SURROUNDINGS

Learning Objectives

- Students will learn to classify the matter based on their physical and chemical properties.
- Students will learn and understand the physical nature of matter.

Learning Outcomes

- Students will be able to classify the matter on the basis of physical properties as solids, liquids and gases; on the basis of chemical properties as elements, compounds and mixtures.
- Students will be able to explain that the matter is made up of tiny particles.

Real Life Application

A garden is a place where kids can touch and feel the fruits and brightly coloured flowers, smell the natural scents and hear the sound of rustling shrubs.

Kids love to play and spend time in the garden. They enjoy making puddles of water while watering plants, playing in the sand which slips when taken in hand and a lot more.



A garden is one of the best examples where objects of various forms, shapes, sizes and textures can be seen. You must have wondered what these objects are basically made up of or what the building blocks of these objects are. These building blocks are known as 'matter'.

Let us now study and learn about various types of matter which are found in our surroundings.

“Anything which occupies space and has mass is called matter”.

Classification of Matter

Modern-day scientists have concluded that there are two ways for classifying matter based on their physical and chemical properties.

- (1) On the basis of physical properties, the matter is classified as solid, liquid and gas.
- (2) On the basis of chemical properties, the matter is classified as elements, compounds and mixtures.

Physical Nature of Matter

As the matter is a physical entity, it has some physical features. These features can be broadly explained as:

- (1) Matter is made up of tiny particles, which cannot be seen through the naked eyes and can be observed through a microscope.
- (2) Every matter in the universe has these tiny particles as their building blocks.
- (3) As the matter has building blocks, we can say that matter is not continuous but particulate in nature.

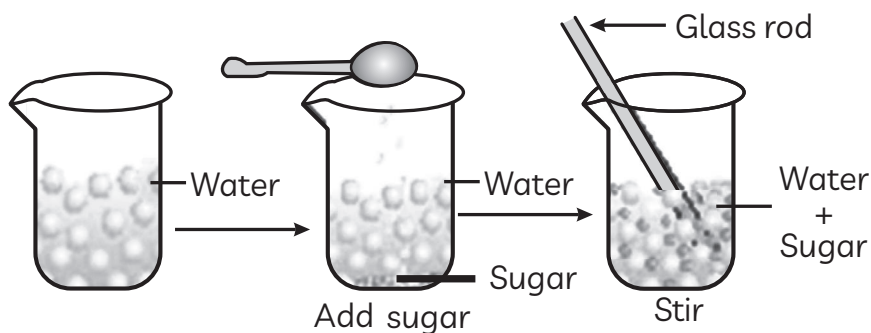
Example 1. Which of the following are matter? chair, air, love, taste, hate, respect, juice, hot, cold drink, the smell of samosa, table and cookies.

Ans. Chair, air, juice, cold drink, table and cookies.

Matter is made up of tiny particles

Matter is made up of very small particles. The particles which make up matter are atoms or molecules. For example, 1 gram of Gold is made up of 3×10^{21} Gold atoms.

Example 2. In the Chemistry lab, the teacher performed an activity. He took 100 mL of a beaker, filled it with 50 mL of water and marked it. He then dissolves some sugar with the help of a glass rod until the sugar crystals can no longer be seen and are completely dissolved.



Particles of water magnified millions of times

- (A) Does the level of water change?
- (B) What has happened to the sugar?
- (C) Where does it disappear?

[NCERT Activity 1.1]

Ans. (A) No, the level of water does not change.

(B) When dissolved in water, the particles of sugar occupied the spaces present between the particles of water.

(C) Water is a liquid and has empty spaces as the particles of water are loosely packed. When sugar is added to water, the particles of sugar crystals get into the empty spaces between particles of water. As a result, the sugar dissolves and appears to us that it has disappeared.

Related Theory

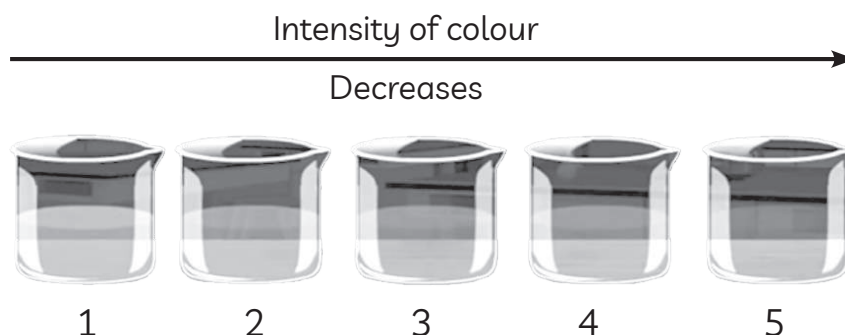
→ If only a small amount of sugar or salt is dissolved in water, the water level does not rise. If we add a large amount of sugar or salt in water, the water level rises.

Example 3. Nina takes a crystal of potassium permanganate but it accidentally falls into the beaker of 100 mL water and get dissolves into it. She takes 10 mL of this solution and dissolves 90 mL of clear water in it. Now, again she takes 10 mL of this diluted solution and dissolves it in another 90 mL of clean water and repeat the process 4-5 times.

- (A) Is the water still coloured?
- (B) What observation did Nina make?
- (C) What did Nina infer from this experiment? **[NCERT Activity 1.2]**

Ans. (A) Yes, the water is still coloured.

(B) The solution held in the various measuring cylinders is coloured, though the colour gets fainter with successive dilutions.



(C) Nina inferred that there are a million of tiny particles even in a few drops of coloured water that keep dividing themselves into smaller and smaller particles.

Example 4. Is matter continuous or particulate? Explain by taking an example.

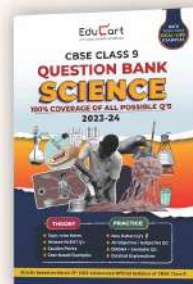
Ans. Matter is not continuous and is particulate in nature, i.e., it is made up of particles. The cricket bat is made up of wood which in turn is made up of tiny particles of wood even though it seems to be one undivided piece. Similarly, sand is also made up of tiny particles.

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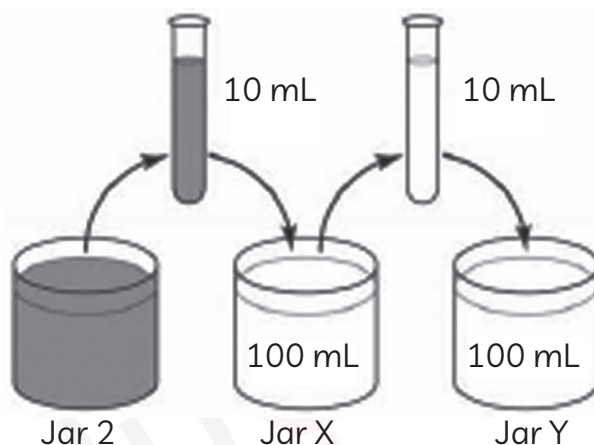


OBJECTIVE Type Questions

Multiple Choice Questions

[1 mark each]

1. Masood took 10 mL of the coloured water from Jar 2 and poured it into 100 mL of plain water in Jar X. He then took 10 mL of the mixture from Jar X and poured it into 100 mL of plain water in Jar Y as shown for the figure.



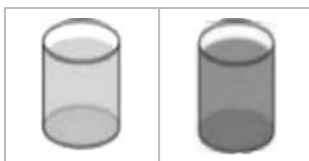
He mixed the liquids in jars X and Y.

What would be the likely colour of the liquids in Jar X and Jar Y respectively?

	Colour of liquid in Jar X	Colour of liquid in Jar Y
(a)		
(b)		
(c)		
(d)		

[CBSE Question Bank 2022]

Ans. (c)



Explanation: The solution present in jars X and Y is coloured though the colour gets fainter and fainter with successive dilution. Masood inferred that there are a million of tiny particles present in jar 2 even in a few drops of coloured water that keep dividing themselves into smaller and smaller particles. It means the particles of matter are very small.

2. The boiling points of diethyl ether, acetone and *n*-butyl alcohol are 35°C, 56°C and 118°C respectively. Which one of the following correctly represents their boiling points in the Kelvin scale?

- (a) 306 K, 329 K, 391 K
- (b) 308 K, 329 K, 392 K
- (c) 308 K, 329 K, 391 K
- (d) 329 K, 392 K, 308 K

[NCERT Exemplar]

Ans. (c) 308 K, 329 K, 391 K

Explanation: Kelvin Temperature = Degree Celsius + 273

The boiling point of diethyl ether, acetone and *n*-butyl alcohol is:

$$35^{\circ}\text{C} + 273 = 308 \text{ K}$$

$$56^{\circ}\text{C} + 273 = 329 \text{ K}$$

$$118^{\circ}\text{C} + 273 = 391 \text{ K}$$

Thus, 308 K, 329 K and 391 K represent their boiling points in the Kelvin scale of temperature.

3. A teacher brings ice in the class. He starts heating it. Which of the following change will take place when the ice will be heated?

- (I) Kinetic energy of particles increases
- (II) Increase in pressure
- (III) Increase in compressibility
- (IV) Interparticle force of attraction increases
- (V) Fluidity increases

Options:

- (a) All statements are correct.
- (b) (I), (II) and (IV) are correct.
- (c) (I), (II), (III) and (V) are correct.
- (d) (II), (III), (IV) and (V) are correct.

Ans. (c) (I), (II), (III) and (V) are correct.

Explanation: When a solid is converted into liquid, particles gain more kinetic energy due to this the compressibility increases as well as the pressure increases. As the spacing is not close, it will not be rigid, it will be easy to flow and hence, will show fluidity

Assertion-Reason (A-R)

[A-R]

[1 mark each]

In the following questions, two statements in the form of an Assertion (A) and a Reason (R) have been put forward. Read both statements carefully and choose the most appropriate option:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

4. Assertion (A): Water freezes at 0 degree Celsius while it boils at 100 degree Celsius.

Reason (R): Water can exist in all three states of matter—solid, liquid or gas.

Ans. (b) Both A and R are true and R is not the correct explanation of A.

Explanation: Water can exist in all three states of matter. It becomes ice at 0 degrees Celsius and evaporates at 100 degrees Celsius.

Fill in the Blanks

5. Latent heat of vaporisation is the heat energy required to change 1 kg of a liquid to gas at atmospheric pressure at its

..... .

Ans. boiling point

Explanation: The heat energy required to convert 1 kg of a liquid to gas at atmospheric pressure at its boiling point is known as the latent heat of vaporisation.



Caution

→ Students should remember that the latent heat of vapourisation changes a liquid into its vapour without raising the temperature.

Match the Columns

6. The non-SI and SI units of some physical quantities are given in column I and column II respectively. Match the units which belong to same physical quantity.

Column I	Column II
(A) Degree Celsius	(i) Kilogram
(B) Centimetre	(ii) Pascal
(C) Gram per centimetre cube	(iii) Metre
(D) Bar	(iv) Kelvin
(E) Milligram	(v) Kilogram per metre cube

Codes:

- (a) (A) – (i), (B) – (iii), (C) – (ii), (D) – (iv), (E) – (v)
(b) (A) – (ii), (B) – (iii), (C) – (v), (D) – (iv), (E) – (i)
(c) (A) – (iv), (B) – (iii), (C) – (v), (D) – (ii), (E) – (i)
(d) (A) – (iii), (B) – (iv), (C) – (v), (D) – (ii), (E) – (i)

[NCERT Exemplar]

Ans. (c) (A) – (iv), (B) – (iii), (C) – (v), (D) – (ii), (E) – (i)

Explanation:

Physical Quantity	Non-SI unit	SI unit
Temperature	Degree Celsius	Kelvin
Length	Centimetre	Metre
Density	Gram per centimetre cube	Kilogram per metre cube
Pressure	Bar	Pascal
Weight	Gram	Kilogram



CASE BASED Questions (CBQs)

[4 & 5 marks each]

Read the following passages and answer the questions that follow:

7. When heat energy is applied to water, particles begin to move faster. When the particles reach a specific temperature, they have enough energy to break the forces of attraction between them. The liquid begins to turn into a gas at this temperature.

(A) What name is given to the temperature at which liquid starts boiling at atmospheric pressure?

(B) Boiling is a bulk phenomenon. Explain?

(C) Is evaporation the same as boiling? Explain.

Ans. (A) The temperature at which the liquid boils and changes into a gaseous state at the atmospheric pressure is called the boiling point.



Important

→ Water boils at 100°C to form water vapour.

(B) Bulk phenomenon is a phenomenon in which every molecule of a bulk gains enough energy due to heating to get converted into vapours.

(C) Evaporation is not the same as boiling because the process of converting a liquid into vapours at any temperature above melting point and below its boiling point is known as evaporation. While boiling is the process of vaporisation at boiling point where vapour pressure becomes equal to atmospheric pressure.

VERY SHORT ANSWER Type Questions (VSA)

[1 mark each]

8. What are the properties of particles of matter?

Ans. The properties of particles of matter are as follows:

- (1) The size of particles of matter is extremely small.
- (2) There is space between particles of matter.
- (3) Particles of matter are attracted to one another.
- (4) Particles of matter are continuously moving.

SHORT ANSWER Type-I Questions (SA-I)

[2 marks each]

9. Give a reason for the following:

(A) Cooling occurs due to evaporation.

(B) The evaporation rate of the aqueous solution decreases as the humidity rises.

Ans. (A) Cooling occurs due to evaporation. This is because when the liquid vaporises, it absorbs the latent heat of vaporisation from the “other substances” that it comes into contact with. These substances lose heat and cool. Hence, the evaporation causes cooling.

(B) If the humidity is high, the air is already saturated with water vapour. Therefore, it is difficult to absorb more water vapour. Therefore, the evaporation rate will decrease.

SHORT ANSWER Type-II Questions (SA-II)

[3 marks each]

10. While cooking food, when we cover it with a lid, we find bubbles on it. What's the reason behind this?

Ans. Boiling water contains two phases: liquid and gas at the same time. On a cooktop at sea level, the water and steam are both 100°C .

The lid on a pot is originally at room temperature and works as a condenser; around 100°C , the steam cools as it loses heat to the lid and returns to a liquid phase.



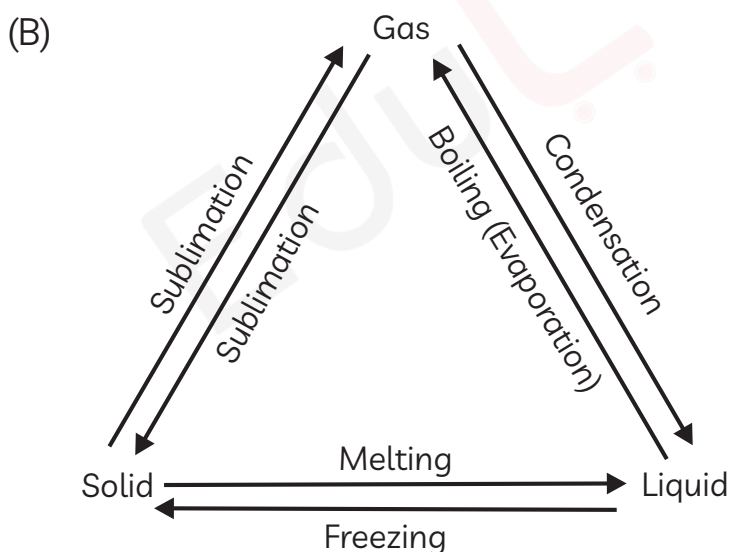
When the lid is first placed on the pot, it is at room temperature, thus steam condenses to water when it comes into contact with the lid; however, the lid quickly warms up, and the water drops on the lid evaporate.

LONG ANSWER Type Questions (LA)

[4 & 5 marks each]

- 11. (A)** What are the two ways in which the physical states of matter can be changed?
- (B)** Draw the 'states of matter triangle' to show the interconversion of states of matter.
- (C)** How can the evaporation of a liquid be made faster?

Ans. (A) The two ways by which the physical state of the matter can be changed is either by melting or boiling.

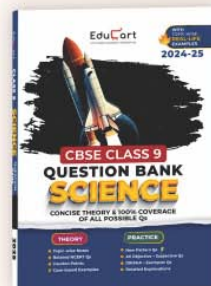


- (C) The evaporation of liquids can be made faster by:
- (1) Increasing the temperature of the liquid
 - (2) Increasing the surface area of liquid
 - (3) Increasing the wind speed



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